MP Online • Internship Project • 2025-26

Expense Tracker with Smart Insighter

Requirements Traceability Matrix (RTM)

Mapping Requirements to Design and Test Cases

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MP Online Internship Project

1. Overview:

Requirement Traceability Matrix (RTM)

The Requirement Traceability Matrix (RTM) is used to ensure that all system requirements are properly implemented and tested. It establishes a clear relationship between functional requirements, system modules, implementation details, and corresponding test cases.

The RTM helps in:

- Verifying that all requirements are covered.
- Ensuring complete testing of system features.
- Improving traceability and project quality.
- Identifying missing or untested functionalities.

2. Traceability Matrix:

Req ID	Requirement Description	Module / Feature	Implementation (API / Logic)	Test Case ID	Priority	Status
FR1	User Registration	Authentication Module	POST /register	TC03, TC04	High	Passed
FR2	User Login	Authentication Module	POST /login	TC01, TC02	High	Passed
FR3	Add Expense	Expense Management	POST /expenses	TC05, TC06	High	Passed
FR4	View Expenses	Dashboard	GET /expenses	TC14, TC15	High	Passed
FR5	Edit Expense	Expense Management	PUT /expenses/{id}	TC07	Medium	Passed

Req ID	Requirement Description	Module / Feature	Implementation (API / Logic)	Test Case ID	Priority	Status
FR6	Delete Expense	Expense Management	DELETE /expenses/{id}	TC08	Medium	Passed
FR7	Automatic Categorization	AI / Logic Module	Rule-based NLP Service	TC09	High	Passed
FR8	Budget Setting	Budget Module	POST /budget	TC10	High	Passed
FR9	Overspending Alert	Alert System	Backend Alert Logic	TC11	High	Passed
FR10	Predictive Analysis	AI / Logic Module	Prediction Logic (Average-based)	TC12	Medium	Passed
FR11	Anomaly Detection	AI / Logic Module	Statistical Deviation Logic	TC13	Medium	Passed
FR12	Data Visualization	Dashboard UI	Recharts Library (Pie & Line Charts)	TC14, TC15	Medium	Passed
FR13	Logout Functionality	Authentication Module	Frontend Session Handling	TC16	High	Passed
FR14	Route Protection	Security / Auth	Protected Routes Logic	TC17	High	Passed

3. Component to Requirement Mapping:

This table maps each component or file to the functional requirements it satisfies, ensuring complete traceability between implementation and system requirements.

Component / File	Type	Satisfies Requirements
user_model.py	Backend - DB Model	FR1, FR2
expense_model.py	Backend - DB Model	FR3, FR4, FR5, FR6
auth_service.py	Backend - Service	FR1, FR2, FR13, FR14
expense_service.py	Backend - Service	FR3, FR5, FR6
ai_service.py	Backend - Service	FR7, FR10, FR11
budget_service.py	Backend - Service	FR8, FR9
auth_router.py	Backend - Router	FR1, FR2
expense_router.py	Backend - Router	FR3, FR4, FR5, FR6
budget_router.py	Backend - Router	FR8
ai_router.py	Backend - Router	FR7, FR10, FR11
dependencies.py / security.py	Backend - Core	FR2, FR13, FR14
Login.jsx + Register.jsx	Frontend - Pages	FR1, FR2
Dashboard.jsx	Frontend - Pages	FR4, FR12
AddExpense.jsx	Frontend - Pages	FR3
ExpenseList.jsx	Frontend - Pages	FR4, FR5, FR6
Charts.jsx	Frontend - Components	FR12
Alerts.jsx	Frontend - Components	FR9
API Service (Axios)	Frontend - Service	FR1 – FR14